

LEDGER · 2026-08-27 · MATERIALS SCIENCE / CHEMICAL ENGINEERING

High-Purity Mesoporous Silica Nanoparticles via Alkaline Extraction of Rice Husk Ash

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	SYLOID® 244 FP (W.R. Grace & Co.)
Entry application	Pharmaceutical and Nutraceutical Glidant and Tabletting Excipient
Measured advantage	Parity on purity ($\geq 99.0\%$) and particle size (5-7 μm); price ratio at lower bound is 1.93x (\$6.00 vs \$11.61/kg), falling below the 2x step-change bar
Time to first revenue	24 months
Gross margin at parity	78.0%
Startup CapEx	\$185,000
Payback from first sale	8.5 months

Capital productivity

2.92x

Regulatory risk

Moderate — qualification costs reported (\$300,000 estimated)

Why it failed

- Voided by: Capital and time to revenue, Negative control / sub-2x, Form-factor / handling regression

Assessment

The venture fails velocity due to an estimated 24-month timeline required for pharmaceutical qualification and USP/GMP compliance. It also fails superiority as its performance is only at parity with the incumbent (relying on price/sustainability advantages) and suffers from an inferior dusting form factor.

Also flagged

- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-27 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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